

Agentic AI Immersion Workshop

Pre-Requisites Datasheet • Complete before Day 1

This workshop is hands-on from the first notebook. Every lab authenticates against a live Microsoft Foundry project using your own Azure identity — there are no shared demo tenants and no API keys. Complete every item below **before Day 1** so class time is spent building agents, not troubleshooting environments.

■ Azure role assignments can take 5–15 minutes to propagate. Finish setup at least one day before the workshop, not the morning of.

1. Choose Your Setup Path

Recommended — Dev Container: Install Docker Desktop and Visual Studio Code with the *Dev Containers* extension, clone the repository, then **F1 → Dev Containers: Reopen in Container**. Python, pinned packages, Azure CLI, Azure Developer CLI (azd), and Jupyter support are pre-installed.

Alternative — Local Setup: Install the tools listed in Section 2 yourself and run `pip install -r requirements.txt` in a virtual environment.

2. Required Tools & Accounts

Requirement	Details
Visual Studio Code	Latest version, with the Python, Jupyter, and GitHub Copilot extensions installed.
Git & GitHub account	Git installed locally and signed in (<code>git config / credential manager</code>). A GitHub account is required to clone the repo and use GitHub Copilot; run a test <code>git clone</code> before Day 1 to confirm access.
Python 3.12+ (3.14 recommended)	Matches the pinned dependency lock used by the Dev Container. Confirm with <code>python --version</code> .
Azure CLI	Installed and signed in via <code>az login</code> . Confirm the correct subscription with <code>az account show</code> .
Azure subscription access	An Azure subscription (or resource group) where you can create/access a Microsoft Foundry resource. Contributor-level access is easiest; otherwise request the RBAC roles in Section 4 from your Azure admin in advance.
Docker Desktop	Required for the Dev Container path, and to run a local Redis container used by one Agent Framework lab (<code>threads/2</code>).

3. Microsoft Foundry Project & Model Deployments

Every notebook authenticates with `DefaultAzureCredential` against a Foundry project endpoint — **no API keys are used**. Before Day 1:

- Create a Foundry resource at `ai.azure.com` → **Create project** (this creates the account, a project, and default storage). A region such as East US 2 is recommended for model availability.
- Deploy the following models under **Models + endpoints**: `gpt-5.4`, `gpt-5.4-mini`, `gpt-5.4-nano`, and `text-embedding-3-large`. Wait for each to reach status *Succeeded*.
- Add **Azure AI Search** and **Application Insights** connections under the project's **Connections** tab (used by the search and observability labs).
- Copy `.env.example` to `.env` at the repo root and fill in your project endpoint, model names, and search endpoint.

4. Required Azure Permissions (RBAC)

Ask your Azure administrator to assign the following **data-plane roles** before Day 1 (roles can take 5–15 minutes to propagate after assignment). The repo includes `scripts/setup-permissions.ps1`, an idempotent script that assigns all of these automatically if you have sufficient rights.

Role	Assignee	Scope
Foundry User	Your user account	Foundry account
Cognitive Services OpenAI User	Your user account	Foundry account
Storage Blob Data Contributor	Your user account	Project storage account
Search Index Data Contributor	Your user account	Azure AI Search resource
Search Index Data Reader	Your user account	Azure AI Search resource
Search Service Contributor	Your user account	Azure AI Search resource

Note: The Foundry project's own managed identity and the AI Search managed identity also need matching roles for evaluation and Foundry IQ labs to work; the setup script assigns these automatically. If you don't have permission to assign roles yourself, share this section with your Azure admin ahead of time.

5. Verify Your Setup

Run the following before Day 1 to confirm everything is connected:

- `pip install -r requirements.txt` completes without errors.
- `az account show` returns the subscription containing your Foundry project.
- The verification snippet in the README (loads `AIProjectClient` against your `FOUNDRY_PROJECT_ENDPOINT`) prints **“Connected to Foundry.”**
- Open `azure-ai-agents/1-basics.ipynb` in VS Code and run the first cell — a clean run means you're ready. A 401/403 means roles are still propagating; wait and retry.

6. Optional — Additional Modules

The items below are **not required for Day 1** but unlock specific advanced modules later in the workshop. Set these up only if your agenda includes these sessions.

Module	Additional Requirement
Hosted Agents (notebook 10, hosted-agents/, AgentOps/)	Azure Developer CLI (azd) installed; permission to create/deploy Foundry hosted-agent container resources in your subscription.
Observability & Evaluations (observability-and-evaluations/)	Application Insights connection added to the Foundry project; public network access enabled on project storage (or private endpoints reachable by Foundry) so evaluation traces and content-safety calls succeed.
Agent Framework threads lab (threads/2)	A local Redis instance: <code>docker run -d --name redis-workshop -p 6379:6379 redis:7-alpine</code> .
Model Router lab (13-model-router.ipynb)	The gpt-5.4-mini and gpt-5.4-nano deployments from Section 3; the lab skips any tier that isn't deployed.
AgentOps (GitOps CI/CD demo)	A GitHub repository with Actions enabled, plus permissions to configure Azure federated credentials for CI/CD deployment.

Repository: github.com/dhangerkapil/agentic-ai-immersion • Questions before Day 1? Open a GitHub issue on the repository. • Updated September 2026.